

1. GUIDING PHILOSOPHY: PRIVACY-FIRST & THE ESCALATION BOUNDARY

Coming from an AI Systems and secure multi-tenant infrastructure background, my core architectural principle is that **client PII and sensitive compensation data must never leave the client perimeter to third-party LLMs**. Even when using local open-weight models, AI should strictly act as an advisory co-pilot: it suggests statistically optimal mappings, but final decisions with business liability require human sign-off.

Boundary Rule: Automate transformations that are reversible, verifiable by code, or low-blast-radius; escalate decisions that are irreversible, high-blast-radius, or genuinely ambiguous.

SITUATION / PATTERN	AGENT ACTION	DECISION	ARCHITECTURAL RATIONALE
Whitespace, Case, Phone (+91)	Auto-normalizes to E.164	AUTOMATE	100% reversible, regex-verified, zero data loss.
Date Format (Any day > 12)	Auto-resolves column format	AUTOMATE	Single value (e.g. 24/03/2021) mathematically proves DD/MM format.
Date Format (All days ≤ 12)	Holds format for human review	ESCALATE	04/05/2021 is ambiguous. Guessing permanently corrupts hire dates.
High-Confidence Header Match	Auto-maps to Darwinbox schema	AUTOMATE	Statistically distinct score (≥ 0.75, margin ≥ 0.20) across evidence.
Ambiguous Column (ref)	Surfaces top-2 with LLM context	ESCALATE	Tie between employee_id vs manager_id; risks mangling hierarchy.
Sensitive Fields (Salary, DOB)	Batched 1-click confirmation	ESCALATE	Load-bearing fields require human sign-off once; remembered permanently.
Exact Multi-File Duplicate	Auto-merges & fills null gaps	AUTOMATE	Corroborating identity; non-conflicting values enrich profile safely.
Near-Duplicate (Case 4)	Flags high name match + same DOB	ESCALATE	Merging two distinct individuals into one target ID is irreversible.
Salary Conflict (HRIS vs Payroll)	Withholds auto-merge	ESCALATE	Discrepancy (£80k vs £88k) cannot be auto-picked without legal liability.
Push 503 Transient Failure	Exponential retry with jitter	AUTOMATE	Network/service blip; succeeds silently without polluting review queue.
Target Outage (Repeated 5xx)	Circuit breaker + Batch Rollback	ESCALATE	Fires compensating DELETE calls for active batch to avoid dirty state.

2. DELTA SOLUTIONING: WHY I BUILT “RESOLUTION MEMORY”

The true differentiator of a Forward Deployed Engineer is not calling an LLM API, but engineering the operational delta around it. In enterprise migrations, re-asking consultants to review the same ambiguities on recurring exports causes review fatigue. In DarwinSync AI, when a consultant resolves an escalation (e.g. mapping LOA → On Leave or ref → employee_id) and selects *Remember as rule*, the decision is fingerprinted into the client-scoped rules table.

On re-runs, the agent queries resolution memory *before* raising escalations, creating an active learning loop:

7 → 3
ESCALATIONS OPENED

-57%
HUMAN REVIEW EFFORT

88.5% → 95.1%
AUTONOMY SCORE

100%
AUDIT REPRODUCIBILITY

3. PRODUCTION ROADMAP: WHAT I WOULD BUILD NEXT

- Topological Dependency Graph:** Sequence target entity pushes (Departments → Cost Centers → Managers → Employees) to prevent foreign-key reference rejections during API ingestion.
- Cross-Tenant Synonym Embeddings:** Cluster anonymized header embeddings across enterprise clients to suggest pre-warmed synonym maps while maintaining strict tenant isolation.
- Automated Schema Drift Detection:** Monitor recurring client SFTP drops; automatically detect newly added columns or shifted date conventions and generate draft migration proposals.
- Compensating Saga Orchestration:** Upgrade batch rollbacks to distributed sagas for multi-entity transactional purity.